

CLUJ NAPOCA, Romania

WMO: 151200

Lat: 46.7778N

Lon: 23.5714E

Elev: 411

StdP: 96.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-11.1	-16.2	1.0	-12.5	-13.7	1.2	-10.1	8.9	-0.6	7.3	-0.1	1.2	50	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.4	20.5	29.8	20.0	28.1	19.2	21.9	28.7	20.9	27.6	20.1	26.2	2.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	15.3	24.6	18.9	14.4	23.5	18.0	13.6	22.3	65.8	28.8	62.4	27.5	59.4	26.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.2	5.9	4.8	DB	-17.8	34.1	3.5	2.1	-20.3	35.6	-22.4	36.8	-24.3	38.0	-26.8	39.5
(5)				WB	-17.9	23.7	3.5	1.1	-20.4	24.5	-22.4	25.1	-24.4	25.7	-26.9	26.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.3	0.1	4.8	10.6	15.2	18.7	20.4	20.3	15.4	9.9	4.4	-1.1
	DBStd	8.97	4.61	4.93	4.53	3.84	3.42	3.22	2.85	2.94	3.55	4.04	4.45	4.63
	HDD10.0	1439	380	277	168	38	3	0	0	0	2	53	174	343
	HDD18.3	3350	638	511	421	233	105	33	13	14	99	262	418	601
	CDD10.0	1349	0	0	6	56	166	262	322	318	165	49	7	0
	CDD18.3	218	0	0	0	0	9	46	77	74	12	1	0	0
	CDH23.3	2291	0	0	0	17	144	463	754	749	157	7	0	0
	CDH26.7	691	0	0	0	2	19	124	244	265	35	1	0	0
Wind	WSAvg	1.9	1.6	1.8	2.2	2.3	2.1	2.2	2.0	1.9	1.9	1.8	1.7	1.6
Precipitation	PrecAvg	627	30	25	34	50	70	92	86	64	57	46	37	37
	PrecMax	816	53	58	96	121	136	186	173	172	135	101	88	72
	PrecMin	401	9	6	6	8	27	16	17	3	6	2	1	6
	PrecStd	122	13	12	23	26	30	37	39	40	31	28	17	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	13.2	19.8	25.5	28.9	31.8	33.4	34.2	30.3	24.0	18.1	10.8
		MCWB	7.0	8.3	11.4	15.3	17.4	21.3	20.8	20.5	18.4	15.7	12.7	7.8
	2%	DB	7.0	10.8	16.9	22.6	26.6	29.8	31.1	31.8	27.1	21.6	15.7	8.1
		MCWB	5.2	7.2	9.6	13.9	17.1	20.2	20.7	20.4	17.3	14.7	11.4	6.5
	5%	DB	5.1	8.6	14.3	20.1	25.0	28.0	29.4	29.8	24.9	19.2	13.5	6.2
		MCWB	3.7	5.9	8.3	12.3	16.5	19.3	20.1	20.0	16.8	13.8	10.2	5.0
	10%	DB	3.4	6.6	12.1	18.1	22.9	26.1	27.6	27.6	22.8	16.8	11.1	4.8
		MCWB	2.4	4.7	7.4	11.4	15.8	18.6	19.5	19.1	15.8	12.6	8.8	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.1	12.0	16.2	19.6	23.0	23.0	22.9	20.4	17.1	13.3	8.3
		MCDB	9.0	11.9	17.8	23.5	25.8	29.5	29.3	30.0	26.9	21.8	16.8	9.9
	2%	WB	5.4	7.5	10.3	14.6	18.1	21.3	21.8	21.7	18.6	15.6	11.8	6.5
		MCDB	6.7	10.1	15.4	21.5	24.3	27.5	28.4	28.8	24.5	19.9	14.8	7.9
	5%	WB	3.9	6.2	9.1	13.2	17.1	20.1	20.9	20.7	17.5	14.3	10.5	5.2
		MCDB	4.9	8.3	13.4	18.6	23.1	25.9	27.7	27.4	22.8	18.3	13.0	6.1
	10%	WB	2.5	4.8	7.8	12.0	16.2	19.2	20.0	19.7	16.5	13.0	9.0	4.0
		MCDB	3.3	6.4	11.6	16.6	21.8	24.9	26.2	25.9	21.2	16.0	11.1	4.6
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.7	10.0	11.5	11.9	11.7	12.0	12.6	11.6	10.9	8.0	5.6
		MCDBR	7.4	10.1	14.1	15.1	15.5	14.8	15.1	16.0	15.4	14.6	11.7	7.5
		MCWBR	5.6	6.9	8.0	7.8	7.2	6.6	6.4	6.2	6.7	8.2	7.5	5.7
	5% WB	MCDBR	7.0	9.6	12.8	13.5	13.8	13.3	13.5	14.0	13.2	13.1	10.5	7.2
		MCWBR	5.5	6.8	7.5	7.5	6.9	6.6	6.3	6.2	6.5	7.7	7.1	5.7
Clear-Sky Solar Irradiance	taub	0.312	0.339	0.381	0.426	0.412	0.408	0.427	0.431	0.393	0.365	0.339	0.313	
	taud	2.336	2.290	2.229	2.144	2.245	2.285	2.239	2.226	2.303	2.359	2.380	2.365	
	Ebn at Noon	768	818	833	827	855	858	835	814	814	779	729	721	
	Edh at Noon	75	96	119	142	133	129	133	130	110	89	72	66	
All-Sky Solar Radiation	RadAvg	1.17	1.98	3.15	4.37	5.30	5.83	5.71	5.26	3.72	2.51	1.42	0.91	
	RadStd	0.17	0.26	0.45	0.45	0.51	0.46	0.50	0.49	0.43	0.31	0.20	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.80	N/A	N/A	+1.17	N/A	N/A	-178	-250	+134	+55
(47)	Regional (64 neighbors)	+0.71	N/A	N/A	+0.94	N/A	N/A	-142	-209	+105	+37

Nomenclature: See separate page